MIL-I-46822A(MI)
17 November 1982
SUPERSEDING
MIL-I-46822(MI)
14 September 1964

MILITARY SPECIFICATION

IGNITER, ROCKET MOTOR, XM-69: LOADING, ASSEMBLY AND PACKING

This specification is approved for use by the US Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of the Defense.

1. SCOPE

1.1 Scope. This specification covers the loading, assembly and packing of the XM-69 Rocket Motor Igniter.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-223	Powder, Black.
JAN-P-381	Powder, Propellant 4.2 Inch Chemical Mortar.
MIL-A-2550	Ammunition and Special Weapons, General Specification For.
* MIL-S-46997	Squib, Electric, M3 Manufacture and Assembly.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: DRSMI-RSDS, Redstone Arsenal, AL 35898, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DRAWINGS

Army

8031023	Cartridge, Rocket Motor, XM-69.
8031024	Igniter, Rocket Motor, XM-69.
8031026	Wiring Harness for XM-69 Igniter.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

* 3.1 First Article. Unless otherwise specified (see 6.2), a first article sample of igniters shall meet all the requirements of this specification.

3.2 Materials.

3.2.1 Powder. The powder shall consist of 500 plus or minus 5 grams of class 1 black powder in accordance with MIL-P-223, 225 plus or minus 5 grams of cannon grade black powder in accordance with MIL-P-223, and 275 plus or minus 5 grams of 0.012 TM sheet powder in accordance with JAN-P-381. The sheets shall be cut in 1/8 inch squares. The moisture content of the black powder shall be not greater than 0.7 percent. The material shall be thoroughly mixed before loading. Each igniter shall contain a compact charge of not less than 950 nor greater than 1000 grams.

3.3 Design and Construction. The design and construction of the XM-69 Igniters shall conform in every respect to Drawing 8031024, and as specified herein.

3.3.1 Squibs. The squibs shall conform to MIL-S-46997.

3.3.1.1 Circuits. The squibs shall be connected in series parallel, two circuits in parallel; each parallel circuit shall have two squibs in series. One resistor shall be connected in series with each squib, making a total of four resistors.

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3.3.2 Igniter harness assembly. The igniter harness assembly shall be in accordance with Drawing 8031026.

3.4 Performance and product characteristics.

3.4.1 Resistance. The total electrical resistance of the igniter, including igniter harness assembly and connector, in accordance with Drawing 8031023, shall be 20 plus or minus 2 ohms.

3.4.2 Functioning. The igniter shall fire within 200 milliseconds. No igniter shall show evidence of unburned powder or other malfunction.

3.4.2.1 Current. The electrical current required to fire the igniter, without a delay greater than 200 milliseconds, shall be not greater than 2 amperes. The igniter shall not fire when a current of 18 milliamperes or less, flows through the igniter circuit.

3.4.2.2 Temperature. The igniter shall fire within temperature limits of minus 17.78 degrees Centigrade (C.) (0 degrees Fahrenheit (F.)) to 48.89 degrees C. (120 degrees F.). Each igniter shall operate after storage from minus 28.98 degrees C. (minus 20 degrees F.) to 54.44 degrees C. (130 degrees F.) without subsequent impairment of operation or danger to personnel.

3.5 Workmanship. Uniformity of shapes and dimensions shall permit interchangeability of complete XM-69 igniters. There shall be no cracks, breaks, dents, scratches, loose attaching parts or loose electrical connectors that would adversely affect the igniter for the purpose intended. The igniter shall be clean and free of foreign material.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article. Unless otherwise specified (see 6.2), igniters which have been manufactured under the conditions as those proposed for subsequent units of production shall be submitted for Government approval. The igniters shall be subjected to all the examinations and tests specified herein. Unless otherwise specified the Government will perform the examinations and tests at the suppliers' plant. If the first article sample does not meet all the requirements of this specification, it will be rejected and returned to the supplier. Subsequent units or lots will not be considered for acceptance until approval of the first article sample has been obtained.

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4.3 Lot size. A lot shall consist of 1000 igniters maximum. A smaller quantity may constitute a lot at the discretion of the Government inspector. A maximum of 100 igniters shall constitute the first lot manufactured on any contract except when a new contract immediately follows the manufacture of satisfactory igniters of the same character.

* 4.4 Sampling. The number of igniters to be selected, at random, by the Government inspector from each lot for examination and tests specified herein shall be equal to the number of rocket motor ballistic samples from the lot with which the igniters are to be tested. However, when the igniters are not to be tested in conjunction with the rocket motors, ten igniters shall be selected at random from each lot for examination and tests.

* 4.5 Examination. Examination shall be conducted in accordance with MIL-A-2550 to determine compliance with 3.2, 3.3 and 3.5.

4.6 Tests.

4.6.1 Powder moisture content. The moisture content of the powder shall be determined at intervals as directed by the Government inspector. At least one moisture determination shall be made on the powder used in each lot of igniters. The method for determining the percentage of moisture shall be in accordance with MIL-P-223. If the moisture content is greater than 0.7 percent, the entire lot shall be rejected.

4.6.2 Resistance. Each igniter in the lot shall be tested for resistance by means of a suitable resistance measuring instrument capable of measuring resistance between 0-50 ohms in intervals of 0.25 ohms and having a maximum short circuit of 10 milliamperes. Any igniter which fails to meet the requirements of 3.4.1 shall be rejected.

4.6.3 Functioning. Ten loaded igniter assemblies shall be fired behind a barricade suitable for observing the resulting action. In the event that the igniter assembly and the rocket motor into which it is to be loaded are manufactured by one and the same supplier, the igniter may at the discretion of the procuring activity, be fired in conjunction with the rocket motor ballistic samples. The tests shall be performed under the conditions specified below. The igniters should not fire when a current of 18 milliamperes or less is applied through the igniter leads. The igniters shall fire without a delay greater than 200 milliseconds when a maximum of 2 amperes current is applied through the igniter leads.

4.6.3.1 Elevated temperature. Five loaded igniter assemblies shall be stored at minus 28.98 plus or minus 1.12 degrees C. (minus 20 plus or minus 2 degrees F.) for 24 hours or until an equilibrium temperature within these limits is reached. The igniters shall then be removed and stored at a temperature of 54.44 plus or minus 1.12 degrees C. (130 plus of minus 2 degrees F.) for 24 hours or until an equilibrium temperature within those limits is reached. The temperature shall be lowered to 48.89 plus or minus 1.12 degrees C. (120 plus or minus 2 degrees F.) and tested as specified in 4.6.3. If one igniter fails to meet the requirements of 3.4.2 the entire lot shall be rejected.

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4.6.3.2. Low temperature. Five loaded igniter assemblies shall be stored at 54.44 plus or minus 1.12 degrees C. (130 plus or minus 2 degrees F.) for 24 hours or until an equilibrium temperature within these limits is reached. The igniters shall then be removed and stored at minus 29.98 plus or minus 1.12 degrees C. (minus 20 plus or minus 2 degrees F.) for 24 hours or until an equilibrium temperature within these limits is reached. The temperature shall be raised to 32 plus or minus 1.12 degrees C. (0 plus or minus 2 degrees F.) and tested as specified in 4.6.3. If one igniter fails to meet the requirements of 3.4.2, the entire lot shall be rejected.

5. PACKAGING.

* 5.1 Preservation, packaging, packing, unitization and marking shall be Level A or B (see 6.2).

* 5.1.1 Level A or B. Preservation, packaging, packing, unitization, and marking shall be in accordance with packaging data sheet APN 8031024 and specifications referenced therein.

5.2 Data cards. Data cards shall be furnished in accordance with MIL-A-2550.

6. NOTES

6.1 Intended use. The XM-69 igniter is intended for use as the igniter unit in the booster stags M42A1 Rocket Motor (NIKE HERCULES).

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

* b. Whether or not a first article sample is required.

c. Selection of applicable level of packaging and packing (see 5.1).

* 6.3 Changes to previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Preparing Activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-I-46822A(MI)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

[ ] YES [ ] NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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